

Statistics

Statistics simply means numerical data, and is field of mathematics that generally deals with collection of data, tabulation, and interpretation of numerical data. It is actually a form of mathematical analysis that uses different quantitative models to produce a set of experimental data or studies of real life. It is an area of applied mathematics concern with data collection analysis, interpretation, and presentation. Statistics deals with how data can be used to solve complex problems. Some people consider statistics to be a distinct mathematical science rather than a branch of mathematics. Statistics makes work easy and simple and provides a clear and clean picture of work you do on a regular basis.

Importance of Statistics

Importance of Statistics in Business and Industry

Mathematics and statistics have become ingredients of various decisions problems which is clear from the following:

- (i) In Selecting Alternative Course of Action: The process of business decisions involve the selection of a single action among some set of alternative actions. When there are two or more alternative courses of action, and we need only one course of action, statistical decisions theory helps us in selecting the required course of action by applying Bayesian decision theory and thus saves lot of time.
- (ii) In Removing Uncertainty: In decision-making problems, uncertainty is very common in a situation, when the course of action is not known to us. When there are many possible outcomes of an event, we cannot predict with certainty that what will happen. By applying the concept of joint and conditional probability, the uncertainty about the event can be removed very easily.
- (iii) In Calculating E.O.L., C.O.L., etc.: In business, the opportunity loss is very often, which can be defined as the difference between the highest possible profit for an event and the actual profit obtained for the actual action taken. The expected opportunity loss (E.O.L.) and conditional opportunity loss (C.O.L.) can be easily calculated by using the concept of maximum and minimum criteria of pay-off.

Importance in the Field of Science and Research

Statistics has great significance in the field of physical and natural sciences. It is widely used in verifying scientific laws and phenomenon.

For example, to formulate standards of body temperature, pulse rate, blood pressure, etc. The success of modern computers depends on the conclusions drawn on the basis of statistics.

Importance in the Field of Banking

In banking industry, the bankers have to relate demand deposits, time deposits, credit etc. It is on the basis of data relating to demand and time deposits that the bankers determine the credit policies. The credit policies are based on the theory of probability.

Importance to the State

We know that the subject of statistics originated for helping the ancient rulers in the assessment of their military and economic strength. Gradually its scope was enlarged to tackle other problems relating to political activities of the State.

In the modern era, the role of State has increased and various governments of the world also take care of the welfare of its people. Therefore, these governments require much greater information in the form of numerical figures for the fulfilment of welfare objectives in addition to the efficient running of their administration.

Importance in planning

Planning is indispensable for achieving faster rate of growth through the best use of a nation's resources. It also requires a good deal of statistical data on various aspects of the economy.

One of the aims of planning could be to achieve a specified rate of growth of the economy. Using statistical techniques, it is possible to assess the amounts of various resources available in the economy and accordingly determine whether the specified rate of growth is sustainable or not.

Scope of Statistics

Presents facts in numerical figures

The first function of statistics is to present a given problem in terms of numerical figures. We know that the numerical presentation helps in having a better understanding of nature and of a problem.

Presents complex facts in a simplified form

Generally, a problem to be investigated is represented by a large mass of numerical figures which are very difficult to understand and remember. Using various statistical methods, this large mass of data can be presented in a simplified form.

Studies relationship between two or more phenomena

Statistics can be used to investigate whether two or more phenomena are related. For example, the relationship between income and consumption, demand and supply, etc.

Helps in the formulation of policies

Statistical analysis of data is the starting point in the formulation of policies in various economic, business and government activities. For example, using statistical techniques a firm can know the tastes and preferences of the consumers and decide to make its product accordingly.

Helps in forecasting

The success of planning by the Government or of a business depends to a large extent upon the accuracy of their forecasts. Statistics provides a scientific basis for making such forecasts.

Provides techniques for testing of hypothesis

A hypothesis is a statement about some characteristics of a population (or universe).

Provides techniques for making decisions under uncertainty

Many times we face an uncertain situation where any one of the many alternatives may be adopted. A businessman might face a situation of uncertain investment opportunities in which he can lose or gain.

He may be interested in knowing whether to undertake a particular investment or not. The answer to such problems are provided by the statistical techniques of decision-making under uncertainty.

Limitations of Statistics

Statistics Suits to the Study of Quantitative Data Only

Statistics deals with the study of quantitative data only. By using the methods of statistics, the problems regarding production, income, price, wage, height, weight etc. can be studied. Such characteristics are quantitative in nature.

The characteristics like honesty, goodwill, duty, character, beauty, intelligence, efficiency, integrity etc. are not capable of quantitative measurement and hence cannot be directly dealt with statistical methods. These characteristics are qualitative in nature.

In such type of characteristics, only comparison is possible. The use of statistical methods is limited to quantitative characteristics and those qualitative characteristics which are capable of being expressed numerically.

Statistical Results are not exact

The task of statistical analysis is performed under certain conditions. It is not always possible, rather not advisable, to consider the entire population during statistical investigations. The use of samples is called for in statistical investigations.

And the results obtained by using samples may not be universally true for the entire population. Data collected for a statistical enquiry may not be hundred percent true. Statistical results are true on an average.

Statistics Deals with Aggregates Only

Statistics does not recognize individual items. Consider the statement, "The weight of Mr. X in the college is 70 kg". This statement does not constitute statistical data. Statistical methods are not going to investigate anything about this statement. Whereas, if the weights of all the students of the college are given, the statistical methods may be applied to analyze that data.

According to Tippet, "Statistics is essentially totalitarian because it is not concerned with individual values, but only with classes". Statistics is used to study group characteristics of aggregates.

Statistics is Useful for Experts Only

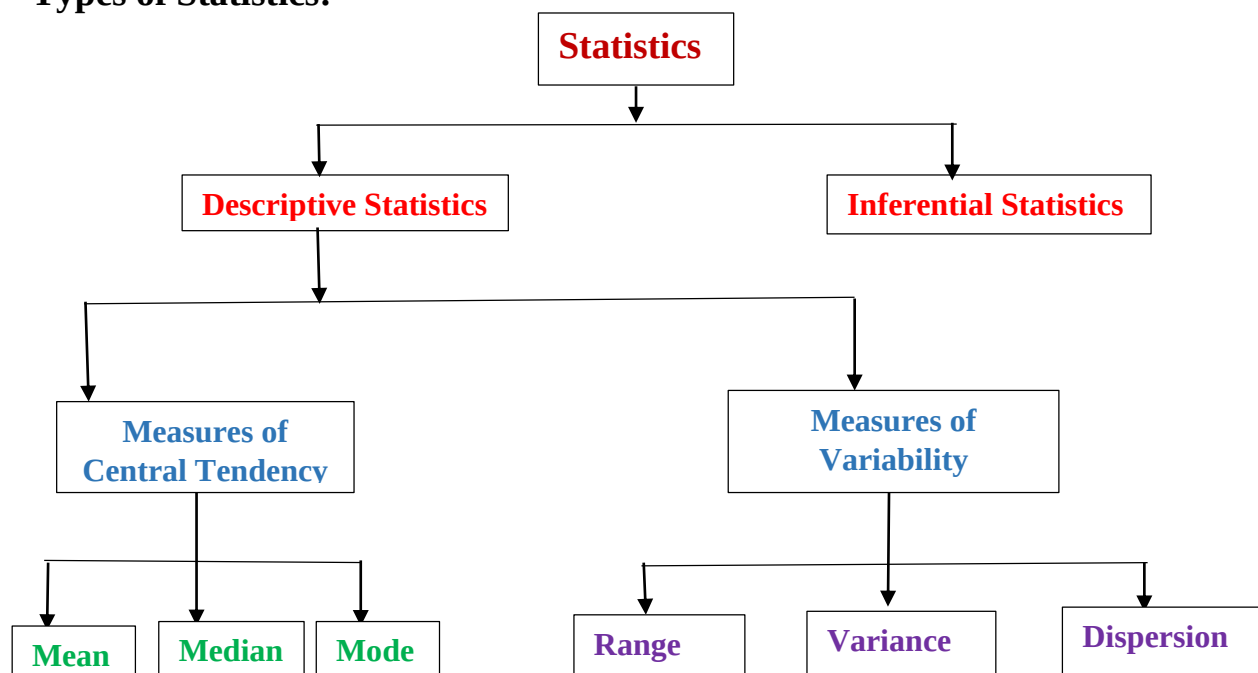
Statistics is both a science and an art. It is systematic and finds applications in studying problems in Economics, Business, Astronomy, Physics, and Medicines etc. Statistical methods are sophisticated in nature. Everyone is not expected to possess the intelligence required to understand and to apply these methods to practical problems. This is the job of an expert, who is well-versed with statistical methods.

Statistics does not Provide Solutions to the Problems

The statistical methods are used to explore the essentials of problems. It does not find use in inventing solutions to problems. For example, the methods of statistics may reveal the fact that the average result of a particular class in a college is deteriorating for the last ten years, i.e., the trend of the result is downward, but statistics cannot provide solution to this problem.

It cannot help in taking remedial steps to improve the result of that class. Statistics should be taken as a means and not as an end. The methods of statistics are used to study the various aspects of the data.

Types of Statistics:



1. **Descriptive Statistics:** Descriptive statistics uses data that provides a description of the population either through numerical calculation or graph or table. It provides a graphical summary of data. It is simply used for summarizing objects, etc. There are two categories in this as following below:

- (a) **Measure of central tendency** Measure of central tendency is also known as summary statistics that is used to represents the center point or a particular value of a data set or sample set.

In statistics, there are three common measures of central tendency as shown below:

(i) Mean (ii) Median and (iii) Mode.

(b) Measure of Variability Measure of Variability is also known as measure of dispersion and used to describe variability in a sample or population. In statistics, there are three common measures of variability as shown below: **(i) Range (ii) Variance and (iii) dispersion.**

2. **Inferential Statistics:** Inferential Statistics makes inference and prediction about population based on a sample of data taken from population. It generalizes a large dataset and applies probabilities to draw a conclusion. It is simply used for explaining meaning of descriptive stats. It is simply used to analyze, interpret result, and draw conclusion. Inferential Statistics is mainly related to and associated with hypothesis testing whose main target is to reject null hypothesis. Hypothesis testing is a type of inferential procedure that takes help of sample data to evaluate and assess credibility of a hypothesis about a population. Inferential statistics are generally used to determine how strong relationship is within sample. But it is very difficult to obtain a population list and draw a random sample.